Multipole decomposition of the density

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I. DEFINITION AND CONVENTIONS

I take the following convention for the spherical harmonics

$$Y_{\ell m}(\theta, \phi) = (-1)^m \sqrt{\frac{(2\ell+1)}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos(\theta)) \mathrm{e}^{\mathrm{i}m\phi}\,, \tag{1}$$

in terms of the associated Legendre polynomials $P_\ell^m(\theta, \phi)$. These are orthonormal w.r.t. integration over the solid angle. This conventions is (i) what Wikipedia calls the "quantum mechanics"-convention, (ii) is what is used in my mean-field code and (iii) is one of (though not the only one!) the most widely used in nuclear physics.
In particular, I note that

$$Y_{00}(\theta, \phi) = \frac{1}{2\sqrt{\pi}}\,. \tag{2}$$

Decomposing an axially symmetric density

$$\rho(r, \theta, \phi) = \sum_{\ell=0}^{\infty} \rho_{\ell 0}(\theta, \phi) Y_{\ell 0}(\theta, \phi)\,, \tag{3}$$

with

$$\rho_{\ell 0}(r) = \int d\theta d\phi \sin(\theta) Y_{\ell 0} \rho(r, \theta, \phi)\,. \tag{4}$$

Care should be taken however with relating these components to observables. For instance, I get

$$A = \sqrt{4\pi} \int_0^{+\infty} dr\, r^2 \rho_{00}(r) \tag{5}$$

where A is the relevant particle number for the density that we decomposed and the prefactor on the r.h.s. is there because of the normalization of Y_{00}. For the **mean** radius squared, I get

$$\langle r^2 \rangle = \sqrt{4\pi} \int_0^{+\infty} dr\, r^4 \rho_{00}(r)\,, \tag{6}$$

with the same prefactor. From this, we can get the **rms** radius from

$$R^2 = \frac{\int d^3r\, r^2 \rho(\mathbf{r})}{\int d^3r\, \rho(\mathbf{r})} = \frac{\langle r^2 \rangle}{A}\,. \tag{7}$$

Taking the convention $Q_{\ell m} = \langle r^\ell Y_{\ell m} \rangle$, I get for the quadrupole moment

$$Q_{20} = \int_0^{+\infty} dr\, r^4 \rho_{20}(r)\,, \tag{8}$$

without a prefactor.

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More generally, I write

$$\rho(r,\theta,\phi) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} \rho_{\ell m}(r) Y_{\ell m}(\theta,\phi) \tag{9}$$

$$= \sum_{\ell=0}^{\infty}\sum_{m=0}^{\ell} \frac{1}{1+\delta_{m0}} \left[\rho_{\ell m}(r) Y_{\ell m}(\theta,\phi) + \rho_{\ell -m}(r) Y_{\ell -m}(\theta,\phi)\right] . \tag{10}$$

Using that $\rho(r,\theta,\phi)$ is a real function and that $Y^*_{\ell m} = (-1)^m Y_{\ell -m}$, one can deduce that

$$\rho^*_{\ell m}(r) = (-1)^m \rho_{\ell -m}(r) , \tag{11}$$

such that we can write

$$\rho(r,\theta,\phi) = \sum_{\ell=0}^{\infty}\sum_{m=0}^{\ell} \frac{1}{1+\delta_{m0}} \left[\rho_{\ell m}(r) Y_{\ell m}(\theta,\phi) + \rho^*_{\ell m}(r) Y^*_{\ell m}(\theta,\phi)\right] , \tag{12}$$

$$= \sum_{\ell=0}^{\infty}\sum_{m=0}^{\ell} (2-\delta_{m0}) \mathrm{Re}\left[\rho_{\ell m}(r) Y_{\ell m}(\theta,\phi)\right] \tag{13}$$

$$= \sum_{\ell=0}^{\infty}\sum_{m=0}^{\ell} (2-\delta_{m0}) \left[\mathrm{Re}\rho_{\ell m}(r)\mathrm{Re}Y_{\ell m}(\theta,\phi) - \mathrm{Im}\rho_{\ell m}(r)\mathrm{Im}Y_{\ell m}(\theta,\phi)\right] . \tag{14}$$

Since I like working with the real and imaginary parts of spherical harmonics, I redefine the expansion coefficients

$$\rho^R_{\ell m}(r) = +(2-\delta_{m0})\mathrm{Re}\rho_{\ell m}(r) = (2-\delta_{m0}) \int d\Omega\, \rho(r,\theta,\phi) \mathrm{Re}\left[Y^*_{\ell m}(\theta,\phi)\right]$$

$$= (2-\delta_{m0}) \int d\Omega\, \rho(r,\theta,\phi) \mathrm{Re}\left[Y_{\ell m}(\theta,\phi)\right] . \tag{15}$$

$$\rho^I_{\ell m}(r) = -(2-\delta_{m0})\mathrm{Im}\rho_{\ell m}(r) = -(2-\delta_{m0}) \int d\Omega\, \rho(r,\theta,\phi) \mathrm{Im}\left[Y^*_{\ell m}(\theta,\phi)\right]$$

$$= (2-\delta_{m0}) \int d\Omega\, \rho(r,\theta,\phi) \mathrm{Im}\left[Y_{\ell m}(\theta,\phi)\right] , \tag{16}$$

where the r.h.s. are now both integrals of real functions and there are no pesky complex conjugations to make things hard. In terms of these, the density becomes

$$\rho(r,\theta,\phi) = \sum_{\ell=0}^{\infty}\sum_{m=0}^{\ell} \rho^R_{\ell m}(r)\mathrm{Re}Y_{\ell m}(\theta,\phi) + \sum_{\ell=0}^{\infty}\sum_{m=0}^{\ell} \rho^I_{\ell m}(r)\mathrm{Im}Y_{\ell m}(\theta,\phi) . \tag{17}$$

Of course, the second set of terms can often be ignored because of the symetries of the nuclear density. Attention, this convention implies again a non-trivial multiplicative factor when calculating multipole moments as in Tantalus/MOCCa

$$\langle Q_{\ell m}\rangle = \int d^3r\, \rho(\mathbf{r}) r^{\ell} \mathrm{Re}Y_{\ell m}(\theta,\phi) \tag{18}$$

$$= \frac{1}{2-\delta_{m0}} \int dr\, r^{\ell+2} \rho_{\ell m}(r) . \tag{19}$$

II. CODE OPERATION AND CONVENTIONS

A. Compilation and execution

The code requires no external libraries and can be compiled by executing

```
make
```

in the main directory. One can configure the Makefile to change the compiler or use the (typical W.R.-style) commands. The resulting `decomposition.exe` executable in the `exec/` folder can be run as

```
./decomposition.exe < input.data
```

This will produce of course a decent amount of STDOUT, but this is all for consistency checks and displaying runtime options; no real "OUTPUT" should be piped into some other file.

B. Input data

The code reads from STDIN (referred to as `input.data` above) a single namelist `main` consisting of the following parameters.

```
NameList /main/ nx,ny,nz,dx, inputfilename, outputfilename, skiprows, &
&               rmin, rmax, nr, maxl, ntheta, nphi, ncol, quantisationaxis
```

```
nx,ny,nz,dx     : parameters of the Lagrange mesh on which the input density is defined.
inputfilename   : filename on which the input density can be read, see below for format
outputfilename  : filename to write the rho_lm components to, see below for format.
skiprows        : how much rows of the input file to skip (because they contain a header, comments, ....)
rmin            : minimum value of r (don't put 0!) in units of fm
rmax            : maximum value of r
nr              : number of points on the r-interval, linearly spaced between
                rmin and rmax (including both end points)
maxl            : maximum l in the components (default = 4)
ntheta, nphi    : maximum number of Gauss-Legendre points in the integrations
                of theta and phi respectively. Should NOT exceed 48!
ncol            : how much densities (i.e. columns) should be read from the input file
quantisationaxis: what principal axis to use for the definition of multipole moments/spherical coordinates
                  1 : X-axis
                  2 : Y-axis
                  3 : Z-axis
                  Note that changing this option will change the output, since you are effectively
                  redefining the spherical harmonics; even if you are not changing the total density.
```

The file specified by `inputfilename` should have the following format

```
Comments/header
X_1  Y_1  Z_1     rho_1(1)   rho_2(1) ...... rho_ncol(1)
X_2  Y_2  Z_2     rho_1(2)   rho_2(2) ...... rho_ncol(2)
                 .......................
X_n  Y_n  Z_n     rho_1(n)   rho_2(n) ...... rho_ncol(n)
```

where x_i, y_i, z_i are the coordinates of point i of the Lagrange mesh (in fm) and $\rho_1(i), \ldots, \rho_{\texttt{ncol}}(i)$ are the values of any number of densities (in fm^{-3}) at this point. The number of densities to read is the parameter `ncol`, while the number of lines the comments/header takes should be given by `skiprows`. Note that you should specify ALL the points on the mesh in this file. It is also entirely YOUR responsibility to give the code the right values of `nx,ny,nz` and `dx`.

C. Output

The code writes something which looks like this to `Outputfilename`:

```
#R [fm]        rho_00 [fm^-3] rho_00 [fm^-3] rho_20 [fm^-3] rho_20 [fm^-3] rho_22 [fm^-3] rho_22 [fm^-3] rho_40 [fm^-3] rho_40 [fm^-3] rho_42 [fm^-3] rho_42 [fm^-3] rho_44 [fm^-3] rho_44 [fm^-3]
0.250000E-02   0.268960E+00   0.261274E+00  -0.316470E-06   0.228053E-06   0.172207E-12  -0.273483E-12  -0.216694E-14   0.525647E-13   0.113895E-15   0.171629E-15   0.135531E-13  -0.424634E-13
0.126475E+00   0.269835E+00   0.260802E+00  -0.803258E-03   0.578254E-03   0.439348E-09  -0.696452E-09  -0.107328E-07   0.337107E-06  -0.513766E-11   0.377150E-11   0.874813E-07  -0.274753E-06
0.250450E+00   0.272357E+00   0.259413E+00  -0.307410E-02   0.220660E-02   0.170205E-08  -0.269128E-08  -0.362077E-07   0.487003E-05  -0.770869E-10   0.568353E-10   0.131230E-05  -0.406304E-05
[......]
0.121520E+02   0.406638E-04   0.104127E-04   0.390502E-04   0.109685E-04  -0.221976E-10  -0.261685E-10   0.166306E-04   0.118469E-04  -0.565681E-10  -0.455869E-10  -0.152837E-05   0.669875E-05
0.122760E+02   0.322171E-04   0.787433E-05   0.298419E-04   0.982551E-05  -0.285844E-10  -0.141093E-10   0.109444E-04   0.108400E-04  -0.440436E-10  -0.440057E-10  -0.165066E-05   0.459245E-05
0.124000E+02   0.258983E-04   0.510765E-05   0.226498E-04   0.820231E-05  -0.341988E-10  -0.146961E-11   0.721794E-05   0.807313E-05  -0.351094E-10  -0.380905E-10  -0.977916E-06   0.173530E-05
```

where the first column specifies the radial coordinate r, `nr` equally spaced points in [`rmin`, `rmax`]. The following columns then contain the components $\rho^R_{\ell m}(r)$, for all $\ell = 0, 2, \ldots,$ `maxl` and all $m = 0, 2, \ldots,$ `maxl`. Each component is repeated `ncol` times, corresponding to the ordering in the original input file.

D. Limitations

Current limitations of this program, although most should be not too hard to change.

- the code does not check your input; it is not foolproof.
- The program is hardcoded to work with EV8/CR8-style spatial symmetries.
- The density being interpolated should occur no signs under any plane reflection symmetry.
- The code only calculates components $\rho^R_{\ell m}$ with ℓ and m both even. (Be aware of the normalisation when $m \neq 0$!)
- The accuracy of the representation of both the original Lagrange mesh and the Gauss-Legendre mesh (θ, ϕ) decreases with increasing ℓ and m. For configurations with small original $Q_{\ell m}$ on the Lagrange mesh, the newly integrated value on the Gauss-Legendre mesh can easily differ by quite a large factor even for large values of `nphi` and `ntheta`. Of course, in terms of $\beta_{\ell m}$ such values are of course entirely negligible...